Observer-Patch Holography and the Dark Sector

Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

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Abstract

Under a supplied universal coupling, the observer-patch Einstein adapter responds to total modular charge, including sources without electromagnetic readout. Two premises, deep-regime scale covariance and quadrature composition of independent sources, characterize a unique one-constant law $a_A = \sqrt{a_b a_0}$ with flat rotation curves and $v^4 = GM_b a_0$, the deep limit of Milgrom's modified dynamics and the baryonic Tully–Fisher relation with a_0 fitted; each premise is load-bearing, the scale covariance is the known symmetry of that deep limit, and the square root reads as the variance additivity of independent collar defects. Anomalous energy on overlap collars is the candidate dark source: under the collar-recovery and modular-stress attachment its stress is bounded by the recovery defect, vanishes at full recovery, and dilutes as a^{-3} when its comoving charge is conserved; a common exponential recovery envelope gives a quantitative compact-source saturation theorem, and the erasure-record model independently computes the optimal recovery defect as $(1-p)/2$. These structural results are machine-checked. A retrospective SPARC comparison on a matched 97-galaxy sample finds the two baryon-subtracted channels consistent with one fitted constant a_0, while the fixed exponent 4 sits 2.66 standard deviations above the ordinary least-squares Tully–Fisher exponent of the seen sample. An independent replay of the empirical full radial-acceleration interpolation authenticates the 153-galaxy parent selection and quantifies its weighting sensitivity.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology pa-

pers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

This paper's hypothesis and place in the tower

This paper sits at layer seven and develops the dark-sector branch. Its hypothesis is that dark matter is non-luminous modular charge: under universal coupling geometry responds to total modular charge, and the anomalous modular energy carried on overlap collars that local records fail to recover is the gravitating dark source, with a deep-regime profile fixed by scale covariance and quadrature composition of independent collar defects.

The finite results are the classification theorems, the bounded source and its cold dilution, compact-source saturation under a common recovery envelope, the erasure-record defect law, the profile characterization, the per-cut dictionary $a_0 = Gn^2c$, and the deep radial-acceleration and Tully–Fisher identities. Each is an algebraic or analytic consequence of the premises named where it is stated and is checked in the Lean formalization [5]. Universal coupling, the recovery envelope, the collar-scale premise and the two profile premises are structure fields or named hypotheses there; the rest-frame Einstein relation and the cap generator split are imported from the spacetime and screen layers [1, 3]. Every physical conclusion is therefore a conditional implication, and the acceleration constant a_0 is a fitted comparison value.

For the branch to apply, the executable model must produce from one native process a cap modular generator with a bulk/collar split, a physical clock and energy normalization in which the collar anomaly is stress in the same dictionary as matter, a responsive three-dimensional geometry whose galactic-scale settled cuts carry an order-one defect, the cut density n and per-cut variance c, a relativistic stress for lensing, and the cosmological abundance. The executable observer [14] supplies none of these: its full configuration reference has a tracial limit with trivial modular flow, and its

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/COMPLETION_PROGRAM.md

native continuum is a two-dimensional screen operator without a radial domain [16, 15]. The branch is refuted by a proof that admitted native repair recovers every collar, since full recovery switches the source off (Theorem 3.3); by a calibrated, prospectively frozen galaxy likelihood that excludes the square-root deep law with one common constant or its Tully–Fisher exponent 4; or by a source derivation of n^2c that returns an a_0 outside the measured range. The frozen register FZ-01 retains an acceleration-scale target from a reserve-survival reading: the presence reading $1.184737388 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, with survival factor $1-P/24$, and the Poisson reading $1.179018696 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, with $e^{-P/24}$. Each reading is scored only against a held-out sub-percent determination of a_0; SPARC and SPARC-derived recalibrations are ineligible, so the fits of Section 7 do not score it, and no verdict is registered [19]. The deep-regime characterization of this paper derives neither reading.

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1 What Sources Gravity in OPH

The conditional Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \tag{1}$$

where $\langle T_{ab} \rangle$ is a supplied local, covariant, conserved stress on the physical branch [1]. Its smooth form requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. Finite positive-null tomography exactly reconstructs a quadratic form from directional charges, but identifying that form with the same local stress additionally requires charge–stress matching, locality and covariance, and

a weak Ward law. In the rest frame the relation then follows from three further premises on a small geodesic ball of radius ℓ: generalized entropy is declared stationary, the bulk entropy variation equals $(8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, and the fixed-volume area variation equals $-(4\pi\ell^4/15)\,\delta G_{00}$. Exact arithmetic gives $\delta G_{00} = 8\pi G\,\delta\langle T_{00}\rangle$.

On that conditional branch, the stress entering this relation is identified with the same modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

2 The Generic Dark Sector

Let a finite family of screen sources be indexed by i, each with rest-frame modular-charge contribution q_i and a flag recording whether it carries the electromagnetic readout. Write Q_{lum} for the sum over flagged sources and Q_{dark} for the sum over the others, so that $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$.

Premise 2.1 (Universal coupling)**.** *The stress contraction entering the rest-frame Einstein relation equals Q_{tot}, with no readout-dependent weight.*

Theorem 2.2 (Geometry responds to all modular charge)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,(Q_{\mathrm{lum}} + Q_{\mathrm{dark}})$.*

Theorem 2.3 (The luminous-only relation)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,Q_{\mathrm{lum}}$ holds if and only if $Q_{\mathrm{dark}} = 0$.*

Corollary 2.4 (A geometric excess forces a dark source)**.** *If $\delta G_{00} \neq 8\pi G\,Q_{\mathrm{lum}}$, some source without the electromagnetic readout carries nonzero modular charge.*

Proposition 2.5 (Attractive sign)**.** *If every non-luminous charge is nonnegative, $Q_{\mathrm{dark}} \geq 0$, and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

Proofs. $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$ is a finite-sum identity. Theorem 2.2 substitutes it into the rest-frame relation. Theorem 2.3 cancels $8\pi G > 0$. The corollary is the contrapositive of the sum of zeros, and the proposition is a sum of nonnegative terms. □

These four lines provide a conditional bookkeeping/classification theorem: if universal coupling holds and a geometric excess over the luminous term exists, that excess is called non-luminous modular charge. They do not derive a nonzero Q_{dark}, its abundance, or a physical carrier. Those existence and attachment questions are separate.

3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\mathrm{anom}} + \mathrm{const}, \tag{2}$$

with B_C the bulk boost generator and K_C^{anom} the nonadditive part carried on the overlap collar [1]. At finite regulator stage the separately supplied collar-recovery model carries this part as a remainder and bounds it:

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \leq C_C\,\eta_\delta, \tag{3}$$

where η_δ is the collar recovery defect and C_C depends only on the collar model and the bounded observable class.

Definition 3.1 (Anomalous stress and density)**.**

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle, \qquad \rho_A := \frac{1}{c^2}\,\delta\langle T_{00}^{\mathrm{anom}}\rangle. \tag{4}$$

The dark-sector adapter identifies this anomalous modular energy with the non-luminous charge of Section 2. Its absence of an electromagnetic readout is a source-localization hypothesis: it is assigned to the part of the modular generator that no local record exposes. Neither identification follows from the finite Einstein-shape theorem. Three consequences follow from (3) and (4).

Theorem 3.2 (Bounded source)**.** $\left|\delta\langle T_{00}^{\mathrm{anom}}\rangle\right| \le 15\,C_C\,\eta_\delta/(8\pi^2\ell^4)$.

Theorem 3.3 (Full recovery switches the source off)**.** *If* $\eta_\delta = 0$ *then* $\delta\langle T_{00}^{\mathrm{anom}}\rangle = 0$. *Along any family with fixed* ℓ*, bounded* C_C*, and* $\eta_\delta \to 0$*, the anomalous stress tends to zero.*

Theorem 3.4 (Cold scaling)**.** *A conserved anomalous charge in a comoving cell of physical volume* $\propto a^3$ *has density* $\propto a^{-3}$*.*

Proofs. Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case $\eta_\delta = 0$ of the bound and a squeeze along the family. Theorem 3.4 is $\rho\,a^3 = Q$. □

The split (2) has a machine-checked finite counterpart. On a finite carrier of microstates labelled bulk or collar, with strictly positive weights, a declared boost datum vanishing on the collar, and modular generator $K = -\log w$, the decomposition $K = 2\pi B + K^{\mathrm{anom}} + c\,\mathbf{1}$ exists with c the bulk-weighted mean of $K - 2\pi B$, is unique among splits whose anomalous part has vanishing bulk mean, and is invariant under the additive-constant quotient $w \mapsto e^{-t}w$. When the bulk weights are exactly Gibbs for $2\pi B$, the anomalous part vanishes on the bulk and is carried entirely on the collar, and its expectation is the remainder bounded in (3). The finite carrier, the boost datum, the normalization clause, and the Gibbs condition are model premises; the receipt is the finite counterpart of (2), at exactly that strength. The executable observer does not supply these premises: its full configuration reference approaches the normalized trace at fixed modular parameter, so it identifies no quantum vacuum and its tracial limit has trivial modular flow, and a bulk/collar split of its source is not constructed [16, 17].

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2\Phi = 4\pi G\,(\rho_b + \rho_A), \tag{5}$$

with the attractive sign of Proposition 2.5 whenever the anomalous charge is nonnegative.

4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect η_δ measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The consensus-layer provenance

package supplies a narrower finite statement [2]: an injected overlap mismatch persists at full strength through any continuation in which no later commit writes a register in that overlap support. This is an unwritten-support persistence theorem. It does not show that a physical galactic cut satisfies the no-write hypothesis or remains long-lived. The proposed physical identification assigns this anomalous record state to galactic collars. The native continuum of one repair history is a two-dimensional spatial operator on the screen with no radial domain, and its internal rank-three modes vanish from bounded-energy families; a bulk reading needs a radial observation algebra that the native energy does not supply [15]. The map from screen collars to nested three-dimensional galactic shells is therefore part of this identification.

Premise 4.1 (Collar scale)**.** *The collars carrying $\delta\langle E_C^{(\delta)}\rangle$ are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density ρ_A of its neighbourhood.*

A conditional saturation theorem supports limiting suppression in the compact-source part of this premise. The collar-recovery claim of the spacetime paper [1] bounds the defect on its conditional mixing branch by an exponential envelope, which we write in the local record density ρ as

$$\eta_\delta \le \kappa\, B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{6}$$

with κ the mixing prefactor, B the boundary size in UV cells, r_0 an offset, ζ the decay constant, and $\delta\rho^{1/3}$ the collar depth counted in UV record spacings.

Theorem 4.2 (Compact-source saturation)**.** *Under the envelope (6), fix common parameters $\kappa \ge 0$, $B \ge 1$, $r_0 \ge 0$, and $\zeta > 0$. Along a family of collars with fixed radius $\ell > 0$ and uniformly bounded nonnegative collar constant C_C, the recovery defect and anomalous stress tend to zero as the depth tends to infinity at fixed positive record density, or as the record density tends to infinity at fixed positive depth. Quantitatively,*

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\, C_C}{8\pi^2\ell^4}\, \kappa B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{7}$$

For $\kappa B > 0$ and $\varepsilon > 0$, the defect envelope bound falls below ε exactly when $\delta\rho^{1/3} > r_0 + \zeta \log(\kappa B/\varepsilon)$. For $\kappa = 0$, the defect and stress vanish at every admissible depth. Thus the bounded stress tends to zero in these fixed-parameter limits.

The delimitation is exact in both directions and machine-checked. The identically zero defect satisfies every envelope hypothesis, so the envelope never forces a nonzero defect: that the galactic-scale settled cuts carry an order-one defect is a premise, not a consequence. And an explicit sparsely recorded instance sits at the bound with an order-one defect and strictly positive stress, so the same hypotheses that screen dense regions admit a nonzero dark source on sparse cuts. What is not established is that physical collars satisfy (6): the envelope is the conditional mixing branch of the collar-recovery claim, taken here as a hypothesis. The executable observer's native repairs have an exact quantum mixing gap $(5 - \sqrt{5})/60$ per uniform seam attempt on the twelve-port carrier [18]; that gap concerns relaxation in native event count and supplies no envelope in collar depth.

The delimitation has a positive counterpart in which the defect is computed rather than admitted. In the erasure-record model, a source bit perfectly correlated with its complement and a

collar record carrying the bit with probability p, the minimum prediction error over all predictors, deterministic or randomized, is exactly $(1-p)/2$: an unrecorded cut carries an order-one defect necessarily, a saturated record carries none, and the defect falls below ε exactly when $p > 1-2\varepsilon$, the mechanism-side counterpart of the settled-cut threshold. With unit collar radius and collar constant, and anomalous energy equal to the model defect, the unrecorded instance carries stress $15/(16\pi^2)$ exactly. The identification of the model defect with η_δ is a modeling premise, and which record density a galactic-scale settled cut realizes is open; the model shows the mechanism class exists, with an exact defect-density law.

Under Premise 4.1 the Solar System is a test of the ambient density alone.

Proposition 4.3 (An isotropic ambient density has no quadrupole)**.** *The potential of a locally uniform density is* $(2\pi G\rho_A/3)\,r^2$*. Its projection on the quadrupole weight* $P_2(\cos\theta)$ *vanishes,* $\int_{-1}^{1}(\mu^2 - \frac{1}{3})\,\mathrm{d}\mu = 0$*, so the quadrupole coefficient* Q_2 *is zero. The only footprint is the isotropic tidal scale* $4\pi G\rho_A/3$*.*

With the local dark density $\rho_A \simeq 6.8\times10^{-22}\,\mathrm{kg\,m^{-3}}$, the value inferred from Galactic kinematics [13], the tidal scale is

$$\frac{4\pi G\rho_A}{3} = 1.9\times10^{-31}\,\mathrm{s^{-2}}, \tag{8}$$

For a dimensional scale comparison, the Cassini trace-free quadrupole result is $Q_2 = (1.6\pm1.8)\times10^{-27}\,\mathrm{s^{-2}}$ [12], four orders of magnitude above the isotropic scale. It is not a direct bound on the traceful isotropic channel, whose quadrupole is identically zero. The anomalous mass enclosed within ten astronomical units is $5\times10^{-15}\,M_\odot$. Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

5 The Deep-Regime Profile

Around a baryonic mass M_b, the proposed collar-count interpretation assigns a number of settled cuts proportional to the enclosing radius r. The precise deep-regime characterization uses the following two premises, each independently falsifiable.

Premise 5.1 (Deep-regime scale covariance)**.** *In the deep regime the enclosed anomalous mass is degree-one homogeneous in the radius:* $M_A(M_b,\lambda r) = \lambda\, M_A(M_b,r)$ *for every* $\lambda > 0$*. A sparsely recorded deep cut retains no length scale, so rescaling the radius rescales the enclosed mass by the same factor.*

Premise 5.2 (Quadrature composition)**.** *The anomalous amplitudes of independent baryonic sources compose in quadrature:* $M_A(m_1+m_2,r)^2 = M_A(m_1,r)^2 + M_A(m_2,r)^2$*. The collar defects of distinct sources are independent, their variances add, and the gravitating anomalous charge is the amplitude of the unrecovered record fluctuation.*

Theorem 5.3 (Profile characterization)**.** *Let* $M_A(M_b,r)$ *be nonnegative, monotone in the source, somewhere nonzero, and satisfy Premises 5.1 and 5.2. Then there is a unique constant* $a_0 > 0$ *with*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \qquad \rho_A(r) = \frac{1}{4\pi r^2}\sqrt{\frac{M_b a_0}{G}}. \tag{9}$$

Proof. Scale covariance factorizes $M_A(M_b, r) = r\, f(M_b)$. Quadrature composition makes f^2 additive on positive sources, monotonicity makes it monotone, and a function additive and monotone on the positive reals is linear: $f^2 = \lambda M_b$ with $\lambda > 0$ by nondegeneracy. Setting $a_0 = \lambda G$ gives (9); uniqueness follows by evaluating at $M_b = r = 1$. Each premise is load-bearing: an additive law $M_A \propto rM_b$ satisfies everything except quadrature composition, and an r^2 law everything except scale covariance. □

Corollary 5.4 (Linear enclosed mass)**.** *The deep-regime density is the shell derivative of the enclosed mass, $M_A'(r) = 4\pi r^2 \rho_A(r)$, with M_A linear in r and the coefficient $\sqrt{M_b a_0/G}$ forced by Theorem 5.3.*

Deep-limit scale invariance is the known symmetry of the modified-dynamics limit [10], and square-root mass laws appear in entropic-gravity proposals [11]. The resulting law is the deep limit of modified Newtonian dynamics, $a_A = \sqrt{a_b a_0}$, with its baryonic Tully–Fisher corollary; the characterization adds that the two premises force it jointly, and it reads the square root as the variance additivity of independent collar defects under source composition. It does not fix a_0.

Premise 5.2 itself follows from a more primitive per-cut model. Suppose the nr nested cuts within radius r each carry anomalous amplitude $\sqrt{V(M_b)}$, the standard deviation of the collar defect sourced by the baryons, with the variances of independent sources adding and the variance monotone in the source. Then the variance function is forced linear, $V(M_b) = cM_b$ with a unique $c > 0$, quadrature composition is a theorem rather than a premise, the induced law satisfies Theorem 5.3, and its constant is exactly

$$a_0 = G\, n^2 c. \tag{10}$$

The dictionary is exact but determines only the product $n^2 c$: two models with different cut densities and equal $n^2 c$ induce the same enclosed-mass law, machine-checked. The magnitude question is therefore the value of the collar density and of the per-cut variance constant; neither is derived here.

One sufficient realization of both premises rests one level deeper. Variance additivity follows from probabilistic independence: for atomic defect contributions with finite second moments, pairwise independence, and a common per-atom variance on one probability space, the variance of the compound defect of every finite atom family is the sum of the per-atom variances, so disjoint-union composition is additive and $V(m) = cm$ on the modeled masses. Independence is stronger than necessary; pairwise zero covariance and other dependent but uncorrelated constructions are viable. An exact linear-asymptotic cut count follows from one rigid geometry: a cut family with a positive first cut, consecutive gaps of at least s, and every length-s interval meeting a cut is exactly the arithmetic progression with spacing s, its count obeys $r/s - 1 \le N(r) \le r/s + 1$, and the counted enclosed mass deviates from the linear law by at most one per-cut amplitude at every radius, with $n = 1/s$. A perfectly correlated pair and a one-cut family show only that the conclusions do not follow after deleting every decorrelation or density condition. They do not make independence or exact spacing necessary. Irregular or ergodic cut families with asymptotic density and sublinear count error are possible. The exact nr per-cut law is the continuum or mean extension; the discrete count above differs from it by a bounded error.

Theorem 5.5 (Deep radial-acceleration relation)**.** *The anomalous acceleration of the profile (9) is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r)\, a_0}, \qquad a_b = \frac{GM_b}{r^2}. \tag{11}$$

Theorem 5.6 (Flat rotation curve and baryonic Tully–Fisher relation)**.** *The circular speed supported by a_A satisfies $v^2 = a_A r = \sqrt{GM_b a_0}$, independent of r, and hence*

$$v^4 = GM_b\, a_0. \tag{12}$$

Proofs. $GM_A/r^2 = G\sqrt{M_b a_0/G}/r = \sqrt{GM_b a_0}/r$, and $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$ for $r > 0$. Multiplying by r removes the radius; squaring gives (12). □

The exponents 1/2 in (11) and 4 in (12) are the parameter-free content of the profile. The constant a_0 collects C_C, the collar density, and the small-ball normalization; none of these is fixed by the source, so a_0 enters every comparison as a fitted value. The deep regime is the range $a_b \ll a_0$, equivalently $r \gg r_M = \sqrt{GM_b/a_0}$. If the deep expression is algebraically extended to the matching radius, both accelerations equal a_0 there and the anomalous term dominates outside r_M; those algebraic statements are machine-checked. The premises are asserted only in the deep regime, so they do not determine the physical transition or the response inside r_M.

6 Lensing and the Relativistic Source

The anomalous charge must enter (1) through one stress tensor; Premise 2.1 supplies no readout-dependent coupling. That fact does not imply equality of masses inferred from null geodesics and timelike orbits. Those observables contract different components of the metric response and can differ in the presence of pressure, anisotropic stress, or gravitational slip. No relativistic refinement limit is supplied that turns the weak-field density (9) into a full T^{anom}_{ab}. A lensing law and the relation between lensing and dynamical mass are therefore physical constructions that this paper does not supply.

7 Comparison with Rotation-Curve Data

Theorems 5.5 and 5.6 are compared with the SPARC sample of 175 disk galaxies with 3.6 μm photometry [7] under the declared inclusive-error deep-law diagnostic: quality flag at most 2, inclination at least 30°, velocity error at most ten percent, and stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges. This leaves 2700 points in 149 galaxies. The strict published-cut replay described below is kept separate. Both computations are released with the paper [8, 6].

Source-bound full-RAR calibration. Separate from the conditional OPH deep law, a deterministic replay fits the published empirical interpolation

$$g_{\mathrm{obs}} = \frac{g_{\mathrm{bar}}}{1 - \exp[-\sqrt{g_{\mathrm{bar}}/a_0}]}$$

to a committed CDS/VizieR snapshot whose provenance, catalogue description, and two table files are pinned by SHA-256. The catalogue cuts reproduce the published 153-galaxy parent selection. The strict point cuts retain 2696 points in 147 contributing galaxies, three more than the published 2693-point census. These counts are distinct from the preceding 2700-point, 149-galaxy deep-law diagnostic: that diagnostic uses the inclusive cut $dV/V \le 0.1$, whereas this full-RAR replay uses the

strict cut $dV/V < 0.1$ and starts from the 153-galaxy parent set. With disk and bulge mass-to-light values 0.5 and 0.7, the unweighted point-level log-residual fit gives

$$a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,\mathrm{m\,s^{-2}}, \qquad \sigma_{\log_{10} g} = 0.1327 \text{ dex},$$

where the width is the standard deviation of a galaxy bootstrap. Equal total weight per contributing galaxy shifts a_0 by -9.58%; weighting only by inverse fractional velocity variance shifts it by $+15.49\%$. These are estimator-sensitivity controls, not calibrated likelihood alternatives. The interpolation and acceleration scale were known before this replay, no OPH source value enters the fit, and distance, inclination, gas, stellar population, intrinsic scatter, and correlation uncertainties are not jointly marginalized. The result certifies public-data custody and arithmetic, not independent evidence for OPH [8, 6].

Radial acceleration. On fixed absolute subsets $g_{\mathrm{bar}} < f a_{\mathrm{ref}}$, with the comparison-only value $a_{\mathrm{ref}} = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ fixed before fitting, the density profile predicts the total acceleration $g_{\mathrm{obs}} = g_{\mathrm{bar}} + \sqrt{g_{\mathrm{bar}} a_0}$. The fixed-exponent diagnostic gives

$$a_0 = 0.843 \times 10^{-10}, \quad 0.875 \times 10^{-10}, \quad 0.924 \times 10^{-10}\ \mathrm{m\,s^{-2}}$$

for $f = 0.3$, 0.1, and 0.03, with 1519, 960, and 203 points in 141, 115, and 40 galaxies and rms scatter between 0.14 and 0.17 dex. Fitting an anomalous power inside the total model, using every retained point rather than selecting on the sign of the noisy excess, gives exponents 0.454, 0.461, and 0.239. Galaxy-cluster bootstrap intervals contain 1/2 for the first two cuts; the deepest cut is sparse and its interval is wide. The computation uses equal retained-point weighting and does not marginalize distance, inclination, stellar-population, or correlated rotation-curve uncertainties, so it is a diagnostic rather than a full measurement likelihood.

Baryonic Tully–Fisher. For the 123 galaxies with a positive measured flat velocity that pass the catalogue quality and inclination cuts (without applying the rotation-point fractional-velocity-error cut), with $M_b = 0.5 L_{[3.6]} + 1.33 M_{\mathrm{HI}}$, the fixed-exponent normalization $a_0 = v_f^4/(GM_b)$ gives $a_0 = 1.55 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ with 0.26 dex scatter, a normalization that carries no row in the retrospective comparison ledger [4], and the free exponent from an ordinary least-squares fit of $\log v_f$ on $\log M_b$ is 3.75 ± 0.10. The exact source exponent 4 sits $+2.66$ standard deviations from that fitted exponent on this seen sample [4].

One constant. Theorems 5.5 and 5.6 share one a_0. A mixed diagnostic compares mismatched observables: the radial-acceleration fit above against the fixed-exponent normalization $a_0 = v_f^4/(GM_b)$ built from the catalogued V_{flat}. It gives paired parent-galaxy bootstrap intervals $[0.730, 1.037] \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $[1.405, 1.723] \times 10^{-10}\,\mathrm{m\,s^{-2}}$ with a log-ratio 95 percent interval of $[0.180, 0.317]$ dex, none of which carries a row in the retrospective comparison ledger [4]. The catalogued V_{flat} is read at a finite measurement radius where the baryonic contribution to V^2 has not been subtracted, and that bias has the direction of the mismatch.

The matched diagnostic removes the mixed observable by comparing the same anomalous observable in both channels: the baryon-subtracted squared speed $v_A^2 = v_{\mathrm{obs}}^2 - v_{\mathrm{bar}}^2$ at the outermost measured radius, restricted to galaxies whose outermost point passes the same fixed absolute deep cut $g_{\mathrm{bar}} < 0.1\, a_{\mathrm{ref}}$ used at $f = 0.1$. Galaxies with any nonzero retained bulge component are excluded because table 1 supplies only total $L_{[3.6]}$, not the disk/bulge luminosity split needed for a consistent mass

denominator. On the resulting 97-galaxy common set the second channel gives $a_0 = 0.769 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, while the radial-acceleration fit on the same common set gives $0.853 \times 10^{-10}\,\mathrm{m\,s^{-2}}$; the paired $\log_{10}$ ratio is -0.045 with galaxy-level bootstrap 95 percent interval $[-0.117, +0.026]$, which contains zero. The full $f = 0.1$ channel-A fit, before the common-set restriction, is $0.875 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. A synthetic sample with a known a_0 recovers a unit paired ratio through the same pipeline, and an independent script re-derives every quoted number from the released receipt. The combined matched-analysis changes remove the displacement and are consistent with a mixed-proxy explanation, but they also change the radius, subtraction, sample, and combination rule together. This does not causally isolate a mixed-proxy explanation; attribution requires staged one-change-at-a-time ablations. The deep-regime profile is consistent with one fitted constant across the matched channels on this seen sample. Consistency on seen data selects nothing: this is a postdiction diagnostic with declared mass-to-light ratios and cuts, and a physical verdict requires the joint nuisance and measurement-error likelihood named in Section 9. The generic modular-charge dark-sector statements do not depend on this profile fit.

Penalized profile objective. A declared-convention objective is released with the paper: a per-point Gaussian rotation-curve data term for $g_{\mathrm{obs}} = g_{\mathrm{bar}} + \sqrt{g_{\mathrm{bar}} a_0}$ with per-galaxy profiled nuisances (disk mass-to-light under a log-normal prior centered at 0.5, distance and inclination at their catalogued errors), and a declared equal-correlation intra-galaxy covariance family scanned over $\rho \in \{0, 0.2, 0.4, 0.6\}$. The likelihood uses the fixed catalogue velocity frame, with mean $v_{\mathrm{model}} \sin i / \sin i_{\mathrm{cat}}$ and catalogue velocity errors. Equivalently, correcting observations and errors by $s(i) = \sin i_{\mathrm{cat}} / \sin i$ requires a density Jacobian $s(i)^n$, which cancels the Gaussian normalization $s(i)^{-n}$. Because the mass-to-light constraint is an astrophysical prior rather than an identified auxiliary-data likelihood, this is a penalized objective, not a maximum-likelihood analysis. Its delta-objective threshold sets are uncalibrated sensitivity contours rather than confidence intervals, and every component is reported. All corrected snapshot threshold sets are connected; a disconnected synthetic curve tests component enumeration. On the $f = 0.1$ deep subset the objective minimum moves from 0.804 to $0.582 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ as ρ runs from 0 to 0.6, a 0.140-dex shift and a large unresolved sensitivity within the scanned covariance family. Other unquantified systematics prevent a total systematic ranking, and no single ρ is selected. The data quadratic form divided by the nominal fixed-nuisance denominator lies between 3.6 and 16.4 on every subset and correlation value, so the declared error family, which carries no intrinsic-scatter term, understates the observed scatter. The paired comparison against the matched baryon-subtracted channel contains zero at every cut and correlation value, but it is only partially paired: channel A is re-minimized in each bootstrap replicate while channel-B nuisance values are held at their full-sample point estimates. The source does not fix a_0; the released receipt records measurement-side sensitivity contours only. They are not acceptance or rejection bands for a future source derivation, which must be scored against a prospectively frozen, calibrated covariance and intrinsic-scatter likelihood. An independent script re-derives the receipt byte for byte, a synthetic sample with known a_0 recovers it through the same pipeline as a regression sanity check, and mutation guards cover the subtraction sign, the cut, the mass-to-light encoding, and the covariance algebra. Independent checks project the model into the fixed observation frame and compare its explicit covariance quadratic form; two exact projected fits at different inclinations must have equal likelihood.

8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, Theorem 3.4 gives $\rho_A \propto a^{-3}$: the anomalous medium has the homogeneous dilution law of a pressureless component. On the sourced causal order of the finite-source cosmology paper the same statement reads: the layered read law in comoving coordinates is the flat order for every scale-factor profile, a comoving cell at layer j has physical volume a_j^3 times its comoving volume, and a layer-independent anomalous charge per comoving cell is the density $\rho_A \propto a_j^{-3}$ by definition of that count measure. This does not derive its pressure perturbations, anisotropic stress, sound speed, or clustering. Premise 4.1 associates its carrier with galactic collars but supplies no perturbation dynamics. The abundance Ω_A, initial data, transfer functions, and the claim that it clusters with galaxies are not derived here.

9 Required Physical Constructions

1. **Collar scale.** Theorem 4.2 supplies asymptotic compact-source suppression under the common exponential envelope (6); exact absence of a compact source's own anomalous charge is part of Premise 4.1. Physical attachment requires a proof that physical collars satisfy the mixing branch behind the envelope, and a positive mechanism for the order-one defect on galactic-scale settled cuts, which the envelope provably does not force.

2. **Magnitude.** The normalization of $\delta\langle E_C^{(\delta)}\rangle$ to repair conservation on the Ward/Bianchi branch is not derived. The per-cut dictionary (10) reduces the question to the collar density n and the per-cut variance constant c: a repair-conservation normalization would fix c, and a source-derived galactic collar count would fix n. On the conditional fixed-capacity branch, reading the screen capacity N as de Sitter horizon entropy in nats gives one acceleration scale without a further dial, the de Sitter value

$$a_{\mathrm{dS}} = c^2\sqrt{\Lambda/3} = \frac{c^2}{\ell_P}\sqrt{\frac{\pi}{N}}, \tag{13}$$

   equal to $5.42 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ at the Planck-central $\Lambda = 1.089 \times 10^{-52}\,\mathrm{m^{-2}}$. That reading of a whole-screen cell count as the area of a three-dimensional region's boundary is a premise; the region map is not constructed [17]. The fitted constant is a fixed fraction of it: $a_0/a_{\mathrm{dS}} = 0.214$ for the full radial-acceleration calibration at disk mass-to-light 0.5, between 0.176 and 0.268 for mass-to-light 0.6 and 0.4, between 0.156 and 0.171 for the deep-regime fixed-exponent fits, and 0.222 for the published $1.20 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. In the dictionary (10) this ratio reads $Gn^2c\,\ell_P\sqrt{N/\pi}/c^2$, so a source derivation of n and c must return a number in that range. The proximity of a_0 to $cH_0/(2\pi)$ is Milgrom's observation [9]; the ratio is recorded as a comparison target and no value of a_0 is derived here.

3. **Profile.** Theorem 5.3 reduces the profile to Premises 5.1 and 5.2; the required step is deriving those two premises from the collar geometry of a point source, and extending the profile through the transition to the Newtonian regime.

4. **Localization.** The finite quotient-invariant receipt of Section 3 identifies the collar operator with the bounded remainder on a finite carrier under the Gibbs-on-bulk condition; the required step is the continuum identification for the physical cap generator, whose modular flow the finite boost datum only declares.

5. **Abundance.** The comoving anomalous charge of the early screen, the resulting Ω_A, and the joint galaxy, cluster, lensing, CMB, and growth likelihood are not derived. Within rotation curves, the released penalized profile objective is a conditional sensitivity diagnostic: a calibrated intra-galaxy covariance, an intrinsic-scatter model, and a source value of a_0 are the measurement-side constructions this paper does not supply.

10 Conclusion

Under universal coupling the screen gravitates according to its total modular charge, so any geometric excess over the luminous term is non-luminous modular charge; this classification does not establish that the excess is nonzero. Anomalous modular energy on overlap collars is a candidate carrier supplied by the collar-recovery, source-localization, stress and conservation premises, not by the finite Einstein-shape theorem. Under those premises it is bounded by the recovery defect, absent at full recovery, and cold when conserved; on galactic-scale collars its Solar-System footprint has zero trace-free quadrupole and a small isotropic tidal scale. Deep-regime scale covariance and quadrature composition characterize its profile, which gives the radial-acceleration relation, flat rotation curves and the baryonic Tully–Fisher relation with one constant. On seen SPARC data the matched channels agree on that constant within the paired bootstrap interval, the penalized profile objective stays consistent at every scanned correlation value with uncalibrated contours, and the source-bound full-RAR replay is a custody and estimator-sensitivity result. A physical verdict requires a calibrated covariance and intrinsic-scatter likelihood; physical application requires attachment of the envelope, a mechanism selecting galactic-scale cuts, and the charge magnitude of Section 9.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Binary erasure channels and optimal prediction in information theory [27].
- Deep-limit scale invariance of modified-inertia phenomenology and square-root mass laws in entropic-gravity proposals [10, 34, 11].
- Exponential decay of conditional mutual information and recovery error in Gibbs mixing regimes [28, 31].
- Modular Hamiltonian splits and boundary-term normalizations in finite systems [35, 23].
- MOND, RAR, LCDM, cluster and CMB dark-sector literature [9, 8, 20, 25, 30].
- packing and counting of separated point sets [26, 22].
- Random-walk and fluctuation-superposition arguments for square-root laws [29, 24].
- RAR and baryonic Tully-Fisher acceleration-scale estimations on SPARC [7, 8, 32, 33].
- SPARC radial-acceleration-relation objective analyses with nuisance profiling [7, 8, 33].
- Variance of independent sums [29, 21].

References

[1] B. Mueller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, OPH spacetime and Einstein paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[2] B. Mueller et al., *Reality as a Consensus Protocol*, OPH consensus paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/reality_as_consensus_protocol.pdf

[3] B. Mueller et al., *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/screen_microphysics_and_observer_synchronization.pdf

[4] B. Mueller et al., *Observer-Patch Holography Cosmology Data and Likelihood Contracts: Provenance, Comparability, and Claim Boundaries*, OPH cosmology data and likelihood contracts paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/cosmology/oph_cosmology_data_likelihood_contracts.pdf

[5] B. Mueller et al., *Observer-Patch Holography: machine-checked Lean formalization*, Einstein-branch modules `SmallBall`, `DarkSector`, `CollarScaleSaturation`, `DeepProfileClosure`, `PerCutCollarComposition`, `SparseRecordDefectWitness`, `CollarPremiseDerivation`, and `FiniteCapGeneratorSplit`, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/Lean/ObserverPatchHolography/EinsteinBranch

[6] B. Mueller, *Deep-regime comparison of the OPH galaxy law with SPARC*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/code/cosmology/rar_deep_regime

[7] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. 152, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/

[8] S. S. McGaugh, F. Lelli, and J. M. Schombert, *Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. 117, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917

[9] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. 270, 365 (1983).

[10] M. Milgrom, *The MOND limit from spacetime scale invariance*, Astrophys. J. 698, 1630 (2009), arXiv:0810.4065. https://arxiv.org/abs/0810.4065

[11] E. Verlinde, *Emergent Gravity and the Dark Universe*, SciPost Physics **2**, article 016 (2017), arXiv:1611.02269. https://arxiv.org/abs/1611.02269

[12] R. S. Park et al., *Improved constraints on modified Newtonian gravity from Cassini radio tracking data*, 2026, arXiv:2602.17884. https://arxiv.org/abs/2602.17884

[13] P. F. de Salas and A. Widmark, *Dark matter local density determination*, Rep. Prog. Phys. 84, 104901 (2021), arXiv:2012.11477. https://arxiv.org/abs/2012.11477

[14] FloatingPragma (2026). The runnable OPH model: observer implementation, tests, receipts and construction guide. https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/model

[15] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/physical_space_time/OCCUPATION_FIELD.md

[16] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/observer_time_fields/REFINEMENT.md

[17] FloatingPragma (2026). Tangent first laws and finite entropy changes: native reference entropy and region boundaries. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[18] FloatingPragma (2026). Native quantum mixing and the full-carrier memory boundary. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/observer/QUANTUM_MIXING.md

[19] FloatingPragma (2026). Frozen prediction ladder. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/FROZEN_PREDICTION_LADDER.md

[20] Aghanim, N., and others (2020). "Planck 2018 results. VI. Cosmological parameters." *Astronomy & Astrophysics* **641**, A6. https://doi.org/10.1051/0004-6361/201833910

[21] Billingsley, P. (1995). *Probability and Measure*. Wiley. ISBN 0-471-00710-2.

[22] Böröczky, J. K. (2004). *Finite Packing and Covering*. Cambridge University Press. https://doi.org/10.1017/CBO9780511546587

[23] Casini, H., Huerta, M., and Myers, R. C. (2011). "Towards a derivation of holographic entanglement entropy." *Journal of High Energy Physics* **2011**, 36. https://doi.org/10.1007/JHEP05(2011)036

[24] Chandrasekhar, S. (1943). "Stochastic Problems in Physics and Astronomy." *Reviews of Modern Physics* **15**, 1–89. https://doi.org/10.1103/RevModPhys.15.1

[25] Clowe, D., Bradač, M., Gonzalez, A. H., Markevitch, M., Randall, S. W., Jones, C., et al. (2006). "A Direct Empirical Proof of the Existence of Dark Matter." *The Astrophysical Journal* **648**, L109–L113. https://doi.org/10.1086/508162

[26] Conway, J. H., and Sloane, N. J. A. (1999). *Sphere Packings, Lattices and Groups*. Springer New York. https://doi.org/10.1007/978-1-4757-6568-7

[27] Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory*. Wiley. https://doi.org/10.1002/047174882X

[28] Fawzi, O., and Renner, R. (2015). "Quantum conditional mutual information and approximate Markov chains." *Communications in Mathematical Physics* **340**, 575–611. https://doi.org/10.1007/s00220-015-2466-x

[29] Feller, W. (1968). *An Introduction to Probability Theory and Its Applications, Volume 1*. Wiley. ISBN 0-471-25708-7.

[30] Hu, W., and Dodelson, S. (2002). "Cosmic microwave background anisotropies." *Annual Review of Astronomy and Astrophysics* **40**, 171–216. https://doi.org/10.1146/annurev.astro.40.060401.093926

[31] Kuwahara, T., Kato, K., and Brandão, F. G. S. L. (2020). "Clustering of conditional mutual information for quantum Gibbs states above a threshold temperature." *Physical Review Letters* **124**, 220601. https://doi.org/10.1103/PhysRevLett.124.220601

[32] Lelli, F., McGaugh, S. S., and Schombert, J. M. (2016). "The small scatter of the baryonic Tully-Fisher relation." *The Astrophysical Journal Letters* **816**, L14. https://doi.org/10.3847/2041-8205/816/1/L14

[33] Li, P., Lelli, F., McGaugh, S., and Schombert, J. (2018). "Fitting the radial acceleration relation to individual SPARC galaxies." *Astronomy & Astrophysics* **615**, A3. https://doi.org/10.1051/0004-6361/201732547

[34] Milgrom, M. (1994). "Dynamics with a nonstandard inertia-acceleration relation: an alternative to dark matter in galactic systems." *Annals of Physics* **229**, 384–415. https://doi.org/10.1006/aphy.1994.1012

[35] Witten, E. (2018). "APS Medal for Exceptional Achievement in Research: Invited article on entanglement properties of quantum field theory." *Reviews of Modern Physics* **90**, 045003. https://doi.org/10.1103/RevModPhys.90.045003

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